

Onboarding Funnel & KYC Drop-Off Optimization

A Data-Driven Product Case Study for a Neobank Onboarding Journey

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Modeled on Indian neobanks (Jupiter, Fi) — mapping the journey from app install to first funded transaction

1. Executive Summary

Of every 10,000 users who install a neobank app, only **1,295 (13.0%) reach a first funded transaction**. The single largest leak in the journey is **Aadhaar Document Upload, where 38.5% of eligible users drop off** — nearly triple the rate of any other stage.

This project maps the full onboarding funnel, isolates *why* users drop off at each stage, and finds two distinct root causes rather than one: a **trust barrier** concentrated among users with a prior complaint or low satisfaction score (54.6% Aadhaar drop-off vs. 36.1% baseline), and a **technical barrier** concentrated among users aged 56+ (19.1% OTP drop-off vs. 9.6% for 18–25). Both barriers collide hardest at the Aadhaar Upload stage, explaining why it is the outsized leak.

The recommended response is a **risk-based KYC policy** (three tracks: Trust-Sensitive, Technical-Risk, and Standard) paired with four prioritized product initiatives — Contextual Trust Messaging, DigiLocker Auto-Fetch, OTP Fallback Channel, and Save & Resume Onboarding — sequenced across a Now/Next/Later roadmap.

Metric	Value
Overall Install → Funded Conversion	13.0%
Worst Single Stage	Aadhaar Document Upload (38.5% drop-off)
Highest-Risk Segment	Prior-Complaint users (3.8% overall completion)
Root Causes Identified	Trust Barrier + Technical Barrier
Initiatives Proposed	4 (RICE-scored, phased roadmap)

2. Problem Statement & Objective

For any neobank, the stretch between app install and first funded transaction is the single highest-leverage phase in the product. It is where acquisition spend either converts into a real customer or is lost entirely, and it is the phase most heavily shaped by regulatory KYC/AML requirements — making it both a growth problem and a compliance problem at once.

Objective

This project set out to answer three questions with data rather than assumption:

- Exactly *where*, stage by stage, do users abandon the onboarding journey?
- *Why* do they abandon — is it one uniform problem, or several distinct ones?
- What specific, prioritized product and policy changes follow from the answer?

3. Methodology & Data

The Funnel

A 10-stage funnel was defined to represent the full onboarding journey: Install → Sign-Up (Mobile Entry) → OTP Verified → Personal Details Entered → PAN Entered → Aadhaar Document Uploaded → Liveness Passed → KYC Success → Bank Account Linked → First Funded Transaction.

Data Source

A public Kaggle dataset (*Customer-Churn-Records.csv*, 10,000 real bank customer records with attributes including Age, CreditScore, Satisfaction Score, Complaint history, and Account Activity) supplied realistic customer profiles. No public dataset tracks granular KYC-funnel-stage events — this data does not exist publicly for any company, since it is commercially sensitive. To bridge this gap, funnel-stage outcomes were **simulated on top of the real customer data**, with each customer's probability of dropping off at each stage weighted by their real attributes (e.g., a prior complaint or low satisfaction score increased the likelihood of dropping off at trust-sensitive stages). Base drop-off rates were grounded in published fintech onboarding benchmarks rather than set arbitrarily.

Stated Assumptions

- The dataset's original *Geography* field (France/Germany/Spain) was relabeled to *Metro Tier 1/2/3* to fit the Indian neobank narrative; underlying distributions were preserved.
- Quarterly install volume of ~50,000 was assumed for sizing impact estimates in Section 8.
- Funnel-stage pass/fail outcomes are simulated, not observed; the customer profiles they are layered on are real.

4. Funnel Analysis: Where Users Drop Off

The table below shows how many of the original 10,000 users reached each stage, and the drop-off rate at each step relative to the stage before it.

Stage	Users Reached	Conversion from Prev.	Drop-off from Prev.
Install	10,000	100.0%	—
Sign-Up (Mobile Entry)	8,086	80.9%	19.1%
OTP Verified	6,812	84.2%	15.8%
Personal Details Entered	5,998	88.1%	11.9%
PAN Entered	4,470	74.5%	25.5%
Aadhaar Document Uploaded	2,748	61.5%	38.5%
Liveness Passed	2,280	83.0%	17.0%
KYC Success	2,055	90.1%	9.9%
Bank Account Linked	1,494	72.7%	27.3%
First Funded Transaction	1,295	86.7%	13.3%

Row highlighted in red: the largest single leak in the funnel.

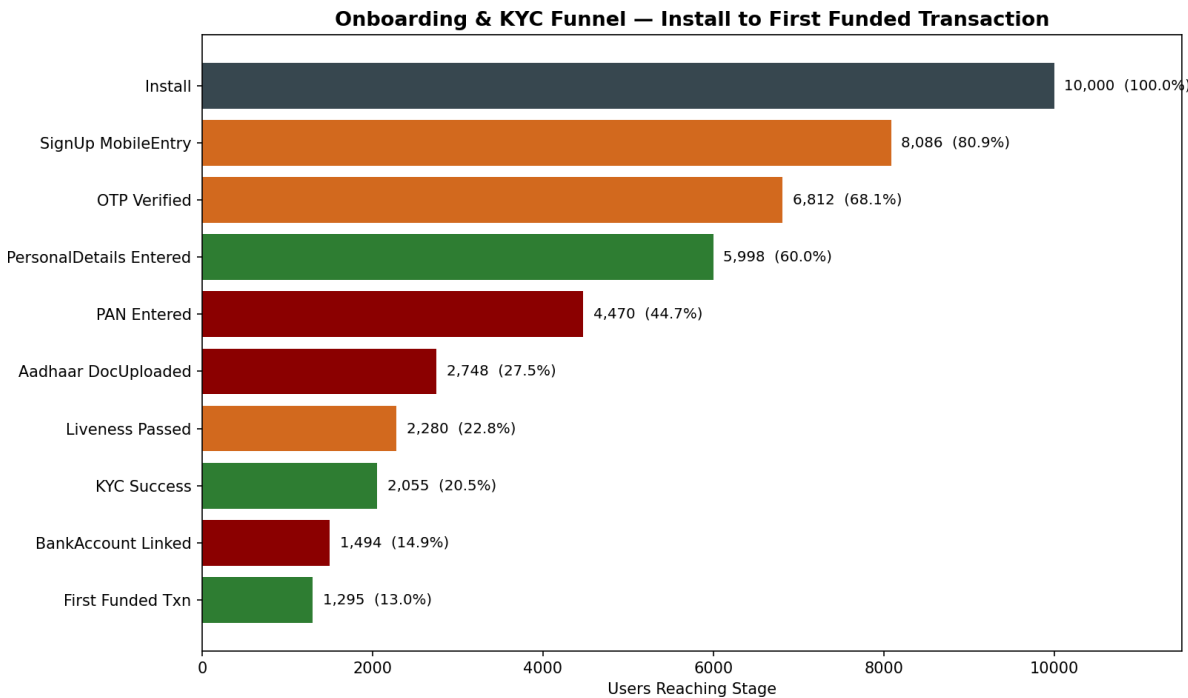


Figure 1. Full onboarding funnel, Install to First Funded Transaction.

Key finding: PAN Entry (25.5%) and Aadhaar Document Upload (38.5%) together account for the two largest single-stage losses in the entire funnel — both are moments where users are asked to share sensitive identity information, the first clue that this is not a purely technical problem.

5. Root Cause: Trust Barrier vs. Technical Barrier

Aggregate funnel numbers show *where* the problem is. Segmenting by real customer attributes shows *why*. Four segments were tested against three representative stages: PAN Entry (a pure trust stage), OTP Verification (a pure technical stage), and Aadhaar Upload (a compound stage combining both).

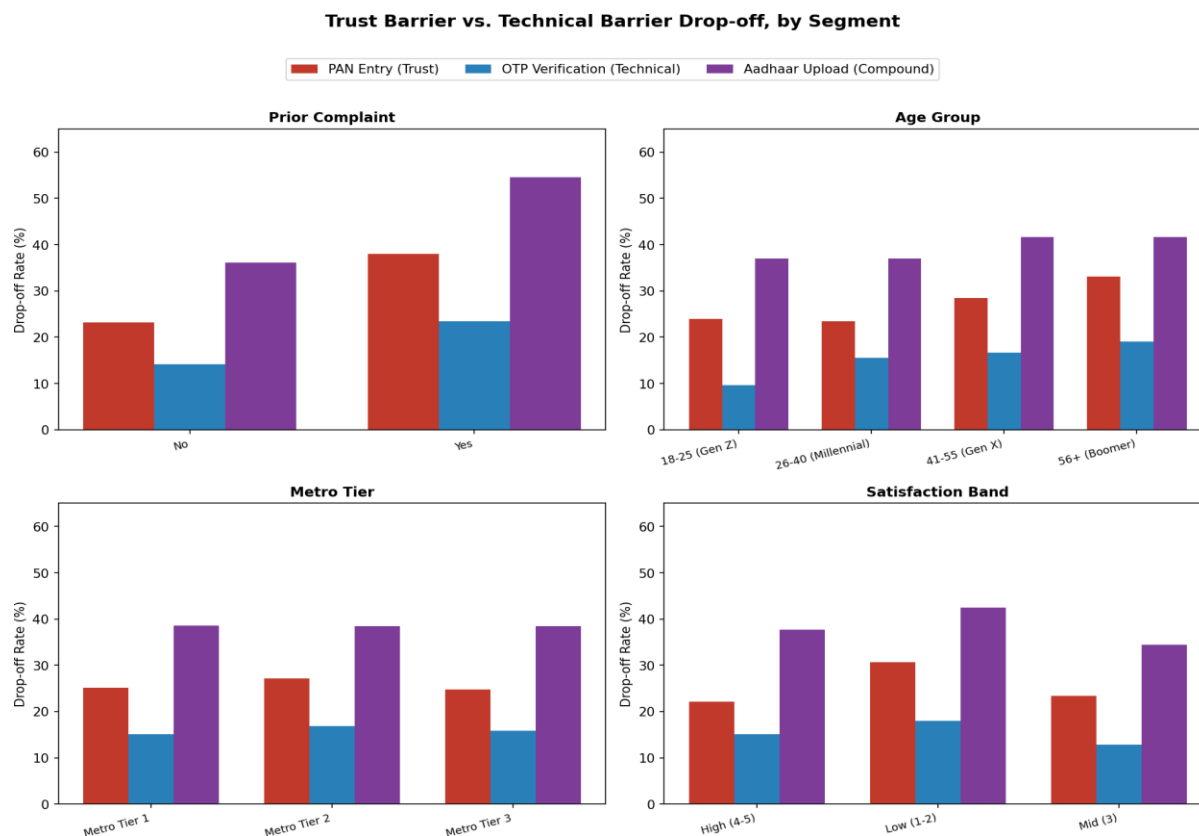


Figure 2. Drop-off rate by segment, split across trust, technical, and compound stages.

Finding 1 — Prior Complaint is the strongest trust-barrier signal

Users with a prior complaint drop off at **38.0%** at PAN Entry (vs. 23.2% baseline) and **54.6%** at Aadhaar Upload (vs. 36.1% baseline) — and complete the full funnel at just **3.8%**, versus 15.3% for everyone else. The effect concentrates specifically at trust-sensitive stages, not evenly across the funnel.

Finding 2 — Age drives the technical-barrier half of the story

Users aged 56+ drop off at **19.1%** at OTP Verification, roughly double the 9.6% rate for 18–25 users — a mechanical/usability effect, distinct from the trust-driven pattern above.

Finding 3 — Metro Tier is not a driver

Completion rates are nearly flat across Metro Tier 1/2/3 (13.0% / 12.5% / 13.4%) — geography was ruled out as a meaningful factor, which sharpens the case that the two segment-driven barriers above are the real story.

Thesis: Onboarding drop-off is not one problem. It is a technical barrier for older users at OTP/Liveness, and a trust barrier for previously-dissatisfied users at PAN/Aadhaar — and the two compound hardest at the single worst stage in the funnel, Aadhaar Document Upload.

6. Proposed KYC Policy

Rather than a single uniform KYC flow, a **risk-based, three-track policy** is proposed. Every track maintains identical verification rigor — the only variable is the communication and support layer around it.

Track A — Trust-Sensitive Users

Triggers: prior complaint on file, or low satisfaction score

- Plain-language trust messaging before PAN/Aadhaar screens.
- Priority human support if a user pauses 60+ seconds on a sensitive-data screen.
- Verification rigor unchanged — intervention is communication-only.

Track B — Technical-Risk Users

Triggers: age 56+, or 2+ failed OTP/upload attempts

- Assisted verification: larger UI text, guided video-call document capture option.
- OTP fallback via voice call after repeated SMS failures.
- Aligned with RBI's existing Video-based Customer Identification Process (V-CIP) — not a new compliance category.

Track C — Standard Users

Everyone else keeps the current self-serve flow unchanged, to avoid slowing down the already-healthy majority.

Governance Baseline (all tracks)

- All KYC decisions remain logged and auditable per PMLA/RBI recordkeeping requirements.
- Risk-track assignment is re-evaluated, not permanent.
- Central KYC Registry (CKYCR) checks apply uniformly across all tracks.

7. Recommended Improvements & Roadmap

Four initiatives were selected and scored using **RICE** ($\text{Reach} \times \text{Impact} \times \text{Confidence} \div \text{Effort}$), then sequenced into a Now/Next/Later roadmap. Sizing assumes ~50,000 installs/quarter.

Initiative	Reach/Qtr	Impact	Confidence	Effort (mo.)	RICE Score
Contextual Trust Messaging	10,000	2	80%	0.5	32,000
DigiLocker Auto-Fetch	30,000	3	70%	3.0	21,000
Save & Resume Onboarding	43,500	1	70%	2.0	15,225
OTP Fallback Channel	6,600	2	90%	1.0	11,880

Trust Messaging tops the list because it is nearly free to build, not because it is the single biggest fix — RICE surfaces quick wins. DigiLocker scores lower purely because it costs 6x the engineering effort, despite addressing the worst leak in the funnel.

Roadmap

Now (Quarter 1) — Fast, cheap, ship immediately

- **Contextual Trust Messaging** — content + minor UI change, no backend dependency.
- **OTP Fallback Channel** — standard voice-OTP integration; directly targets the 56+ technical-barrier stat.

Next (Quarter 2) — The flagship fix

- **DigiLocker Auto-Fetch** — highest overall impact; requires a full quarter for API integration and compliance sign-off.

Later (Quarter 3+) — Safety net, once the flow is stable

- **Save & Resume Onboarding** — most valuable after the core leaks are already reduced, so "resume" catches genuine interruptions rather than masking a broken flow.

8. Expected Impact & Success Metrics

Initiative	Metric Tracked	Current Baseline	Target
Trust Messaging	Aadhaar drop-off, prior-complaint segment	54.6%	Close $\geq 50\%$ of gap to 36.1% baseline
OTP Fallback	OTP drop-off, 56+ segment	19.1%	Within 5 pts of 9.6% baseline
DigiLocker	Aadhaar drop-off, overall	38.5%	Cut by ■ among opted-in users
Save & Resume	7-day resume-to-conversion rate	No baseline	New metric this initiative creates

Each initiative is tracked against its own specific metric rather than overall funnel conversion alone — this is what allows impact to be attributed to a specific change rather than inferred.

9. Tools & Skills Demonstrated

Skill	How It Was Applied
Data Analytics	Python, pandas — funnel construction, weighted simulation, segment analysis
User Experience Design	Diagnosing trust vs. technical friction; designing risk-based flows
KYC/AML Knowledge	RBI Master Direction on KYC, PMLA, V-CIP, CKYCR referenced in policy design
Prioritization Frameworks	RICE scoring, Now/Next/Later roadmapping
Structured Documentation	Policy memo and improvements proposal, written for cross-functional review

10. Limitations & Reflection

This project uses real customer attributes but **simulated funnel-stage events** — no public dataset records granular KYC drop-off, so this is the honest limit of what can be built without access to a live product's analytics. The simulation was deliberately weighted by real customer attributes and grounded in published industry benchmarks rather than left uniform, specifically so the resulting patterns would be plausible rather than arbitrary.

If this were a live product, the first step before building any of the proposed initiatives would be validating the funnel and segment findings against real funnel-tracking tools (Mixpanel or Google Analytics), rather than acting on simulated numbers directly. The KYC policy and improvement recommendations in this document are structured to be directionally correct regardless of the exact underlying numbers — the trust/technical framework and RICE-based sequencing logic would carry over even if the real drop-off percentages differ from the simulation.